

Mamba1 di64 | completed evaluation

Exact original GraphCast loss reduction | 32 matched anchors | 40 six-hour leads

15.80% full-rollout loss reduction

mamba1 di64 | LR 1e-04 | (0.9, 0.98) | BCG4 | swa_step02000-08000

Model / checkpoint	10-day rollout reduction	Day-10 reduction
Mamba1 di64 / selected SWA	15.80%	22.80%
Mamba1 di64 / step4000	13.55%	19.48%
Mamba1 di16 / best rollout SWA	16.28%	23.21%

di64 SWA loss reduction by forecast lead

Day 1: -0.10% | Day 2: 0.02% | Day 4: 3.65% | Day 7: 15.71% | Day 10: 22.80%

di16 comparator: mamba1 di16 | LR 1e-04 | (0.9, 0.98) | BCG1 | swa_step02000-08000

di64 minus di16 rollout reduction: -0.48 percentage points.

Optimizer settings match; BCG count differs, so this does not isolate inner dimension.

Rollout reduction = $100 \times (1 - \text{full_rollout} / \text{baseline_rollout})$, from saved exact losses.

Losses are aggregated over samples and leads before the ratio; day-10 values are separate.

Completed checkpoint + SWA evaluations: 22/24 configurations.

Results use one training seed and an evaluation set used for model selection.

Day-10 GraphCast loss improvement

SWA optimizer matrix | exact loss reduction versus matched frozen GraphCast | higher is better

SWA: 2k-8k updates. * Selected 4k-8k SWA instead.

1 degree / spatial width 512 / all-24-step loss / 10k updates / seed 22 / 32 anchors

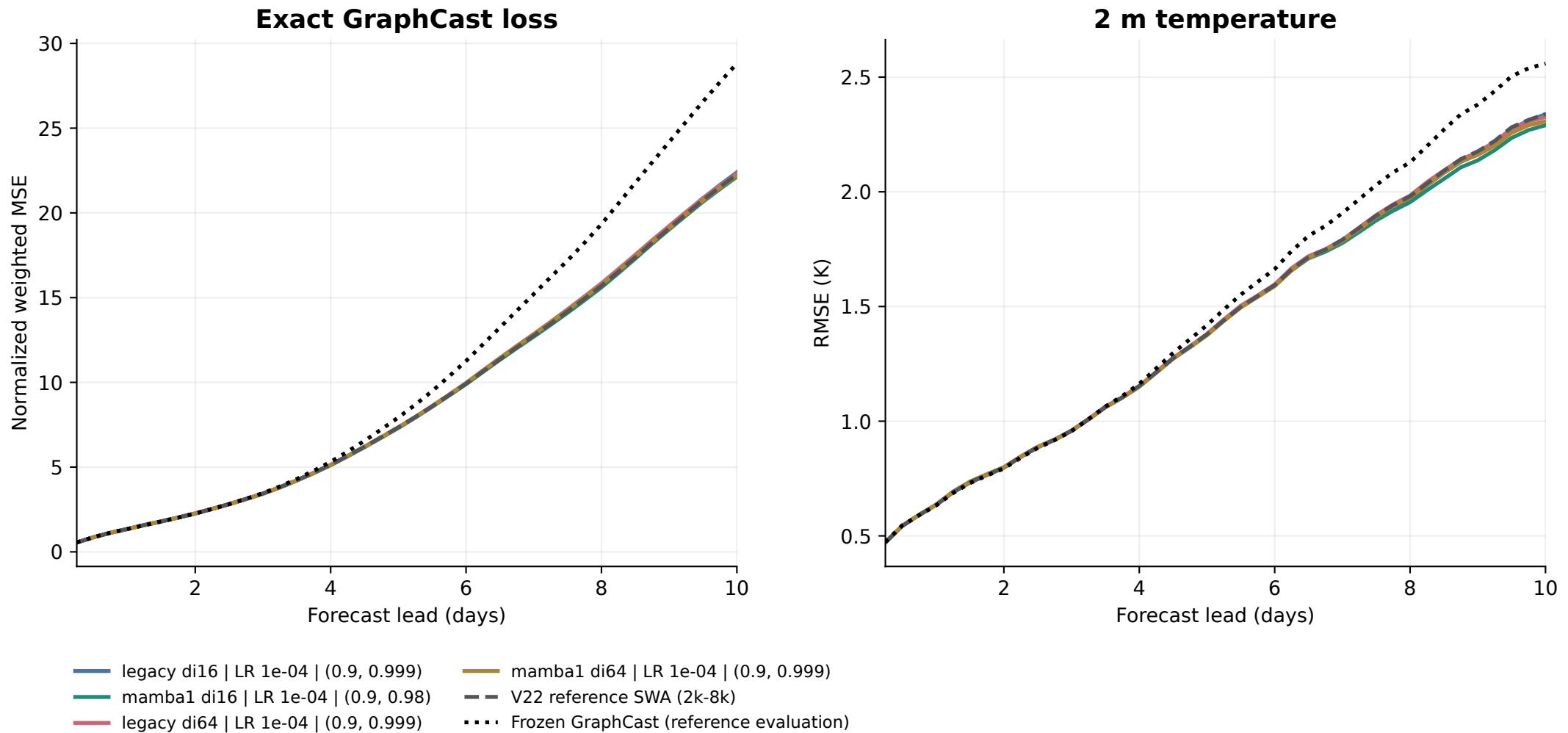
		Peak LR 1e-4			Peak LR 5e-5		
		(0.8, 0.98)	(0.9, 0.98)	(0.9, 0.999)	(0.8, 0.98)	(0.9, 0.98)	(0.9, 0.999)
Legacy	di16 / BCG1	19.62%	20.99%*	22.37%	13.47%	17.40%	16.92%
Mamba1	di16 / BCG1	19.49%	23.21%	22.03%	18.00%	19.38%	20.86%
Legacy	di64 / BCG4	19.34%	20.33%	22.50%	17.20%*	18.98%	21.57%
Mamba1	di64 / BCG4	22.52%	22.80%	23.01%	20.61%	--	--

V22 reference SWA: 22.73% | Frozen GraphCast: 0% | -- awaiting evaluation

Columns show Adam (beta1, beta2). Each cell is 100 x (1 - model loss / matched baseline loss) at day 10. Setup denotes residual-Mamba initialization. This is day-10 improvement, not the rollout-average percentage.

10-day rollout | GraphCast loss and 2 m temperature

Lowest day-10 GC-loss SWA in each initialization/di group; same 32 anchors. Lower is better.

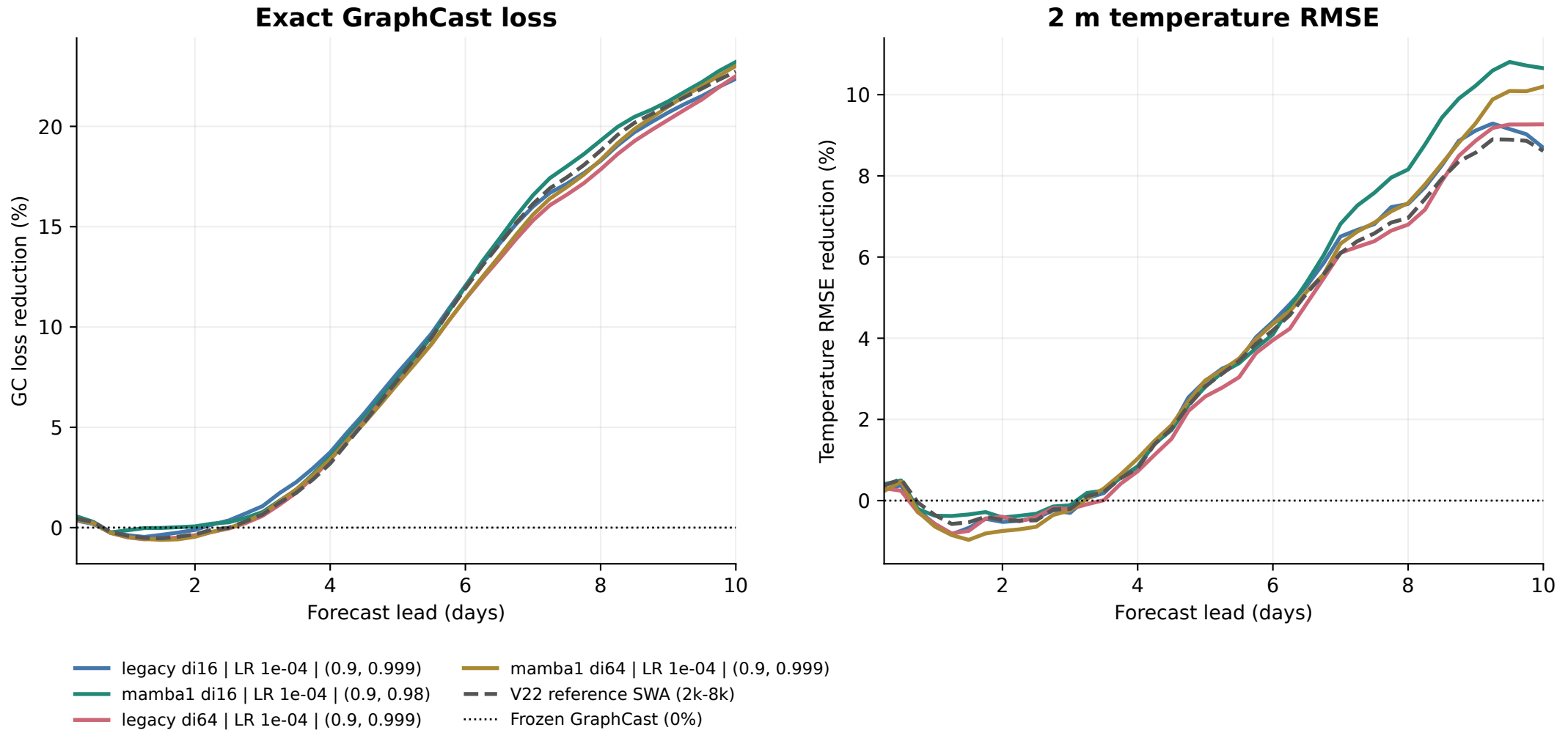


SWA windows: legacy di16: 02000-08000; mamba1 di16: 02000-08000; legacy di64: 02000-08000; mamba1 di64: 02000-08000

GC loss aggregates all forecast variables with the original GraphCast weights. Temperature RMSE is in kelvin. Group winners are selected on these evaluation results; comparisons are descriptive (one training seed).

10-day rollout | improvement over GraphCast

Same SWA models as the absolute-value plots; percentage reduction at each lead. Higher is better.



SWA windows: legacy di16: 02000-08000; mamba1 di16: 02000-08000; legacy di64: 02000-08000; mamba1 di64: 02000-08000

At each lead: $100 \times (1 - \text{model} / \text{baseline})$, using each evaluation's matched frozen baseline.

Left: exact GraphCast loss reduction. Right: temperature RMSE reduction. Negative values mean degradation.

These are per-lead percentages; averaging them does not give the exact rollout-loss reduction.

Evaluation setup and remaining configurations

Details behind the table and rollout curves

Training

Frozen GraphCast Small; residual spatial width 512; two stateful temporal layers.
di16 uses BCG1; di64 uses BCG4. Legacy Haiku and Mamba1 denote initialization schemes.
Closed-loop stop-gradient feedback; segment 96; BPTT 24; AR tail 20; all-24-step loss.
AdamW: 10k updates, 200-update warmup, cosine decay to 0.1 x peak LR; seed 22.

Evaluation and selection

2022 evaluation: 32 matched anchors; cold full-feedback rollout; zero residual state.
40 six-hour leads = 10 days. The matrix reports day-10 GC loss reduction (%).
Each run screens checkpoints 2k/4k/6k/8k/10k and SWA 2k-8k/4k-8k/6k-10k on 8 anchors.
The best checkpoint and SWA by screen rollout-loss reduction are evaluated on 32 anchors.
The curves select the lowest day-10-loss matched SWA within each initialization/di group.
Day-10 frozen-baseline GC loss varies across matrix evaluations: 28.7073 to 28.9940.
The plotted frozen baseline comes from the V22 reference evaluation.

Unfinished configurations (2/24)

mamba1 di64 | LR 5e-05 | (0.9, 0.98): training RUNNING; evaluation not in snapshot
mamba1 di64 | LR 5e-05 | (0.9, 0.999): training RUNNING; evaluation PENDING

Validation and provenance

Checked completed sample counts, matching anchors, baseline checkpoint, evaluation mode,
residual state and resolution; exact GC rollout ratios agree with the saved losses.
Companion CSV includes day-10 GC loss, day-10 temperature RMSE and artifact identities.
Companion sources JSON stores input hashes and the Slurm snapshot. No jobs were launched.